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SR. DATA ENGINEER

Professional Summary

Over 7 + years of extensive experience in Data Engineering, specializing in the design, development, and optimization of data pipelines, data storage solutions, and data processing frameworks for enterprises in various domains such as Finance, Telecom, Retail, and Healthcare.

- Strong expertise in **Big Data technologies**, including **Hadoop Cloudera, Hortonworks, Apache Spark, PySpark, Spark Scala** for distributed data processing and analytics.
- **Proficient in Cloud Technologies** including **AWS S3, EC2, Redshift, Redis, EMR, Kinesis, Lambda, Step Functions, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, IAM, Sage Maker, RDS, API Gateway, AWS CloudWatch, AWS Glue ML Transforms, Amazon Athena.**
- Extensive experience in **Azure Data Lake, Azure Data Factory, Azure Databricks, Azure SQL Database, Azure Synapse, Azure Analysis Services, Azure Blob Storage, Azure Monitor, Azure Key Vault, Azure HDInsight, and Azure Data Explorer** to build scalable, secure, and high-performance **data engineering pipelines, real-time analytics solutions, and enterprise data platforms** that support advanced analytics, machine learning, and business intelligence.
- Proficient in **Google Cloud Storage, Dataflow, Big Query, Pub/Sub, Composer, AI Platform, Cloud Functions, Data plex, and Data Catalog** to design and implement **scalable data pipelines, real-time streaming solutions** enabling efficient data processing and secure storage.
- **Proficient in Python, Scala, and Java** to design and develop scalable, high-performance data pipelines, ETL workflows, and distributed computing solutions using Apache Spark, Hadoop, and cloud-based services.
- **Strong understanding of Azure and Databricks architecture** to design the scalable data solutions utilizing the **Azure Data Lake, Azure Synapse, and Databricks workflows** for advanced analytics and data transformation.
- Expertise in **Snowflake architecture**, optimizing data storage, query performance, and implementing **sharing, multi-cluster warehouses, and role-based access controls RBAC** for enterprise scale analytics.
- Hands-on experience with **SQL and NoSQL databases** such as **Hive, Postgres, MySQL, Oracle, SQL Server, MongoDB, Cassandra, Cosmo DB**, ensuring efficient querying, optimization, and data modeling.
- In-depth knowledge of **file formats** such as **Avro, Parquet, ORC, Text Files, and CSV**, optimizing storage and performance for large-scale data processing.
- Extensive experience with **ETL Tools**, including **Informatica, Azure Data Factory ADF and Ab Initio**, designing and automating data workflows for seamless data integration across systems.
- Skilled in **CI/CD Pipelines**, utilizing **Jenkins, Docker, Kubernetes, and GitLab** for automated deployment, integration, and monitoring of data solutions.
- **Proficient in Version Control Systems** such as **Git, GitLab, GitHub, SVN, and CVS**, ensuring efficient collaboration, code versioning, and CI/CD automation for enterprise data solutions.
- Experienced in **Machine Learning frameworks**, including **AWS Sage Maker, Azure ML Studio, and TensorFlow**, collaborating with data science teams for feature engineering and model development.
- Proficient in **Data Visualization** using **Power BI, Tableau, and Looker**, building insightful dashboards and reports for business stakeholders.
- Hands-on experience in **migrating data from On-premises to Cloud**, such as **Oracle to S3, Teradata to Snowflake**, ensuring smooth data transitions and optimized performance.
- Well-versed in **Agile methodologies (Scrum, Kanban)** using **JIRA, Azure DevOps, Rally, and TFS** for sprint planning, project tracking, and issue resolution.
- Strong understanding of **Databricks architecture**, including **clusters, jobs, notebooks, workspaces, and Delta Lake** for efficient **data processing and storage**.

- Excellent documentation skills using **Confluence, Lucid Chart, and SharePoint** for clear and structured technical documentation, data architecture diagrams, and project reports.

Technical Skills

Languages	Python, Scala, SQL, Java, HTML, CSS, XML
Big Data Tools	Apache Hadoop, Apache Spark, PySpark, Spark Scala, Hive, HBase, Kafka.
Databases	**SQL:** My SQL, Oracle, PL/SQL, PostgreSQL, SQL Server **No SQL:** MongoDB, Cassandra, DynamoDB, Redis, CouchDB
Cloud Technologies	**AWS:** AWS S3, EC2, Redshift, Redis, EMR, Kinesis, Lambda, Step Functions, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, IAM, Sage Maker, RDS, API Gateway, AWS CloudWatch, Aws Glue ML Transforms, Amazon Athena **AZURE:** Azure Data Lake, Azure Data Factory, Azure Databricks, Azure SQL Database, Azure Synapse, Azure Analysis Services, Azure Blob Storage, Azure Monitor, Azure key vault, Azure Inside, Azure Data Explorer. **GCP:** Google Cloud Storage, Dataflow, Big Query, DataProc, Pub/Sub, Composer, AI Platform, Cloud Functions, Data plex, Data Catalog.
BI and Visualization Tools	Tableau, Power BI, Looker
Version Control Tools	Git, GitHub, Bitbucket
Data Integration and ETL Tools	Apache Knife, Azure Data Factory ADF, Informatica, Ab Initio
Orchestration Tools	Apache Airflow, Oozie, Step Functions
CI/CD and Deployment Tools	Jenkins, Docker, Kubernetes
Methodologies	Agile Scrum, SDLC, Continuous Integration/Continuous Deployment
Machine Learning	AWS Sage maker, Azure ML Studio
Data Security and Privacy	Data Masking, Row-Level Security, Column-Level Encryption, GDPR Compliance
Cloud Datawarehouse	snowflake, data bricks

Experience

Capital One **Jan 2023 - Present**

AWS Data Engineer

Responsibilities:

- Designed and implemented scalable cloud-based data pipelines using **AWS Glue**, **Lambda**, and **S3**, enabling seamless transfer of large volumes of financial data from on-premises systems to **AWS cloud storage**.
- Developed and optimized data warehousing solutions using **AWS Redshift**, designing high-performance data models and ensuring efficient querying and reporting capabilities to meet business needs.
- Built distributed data processing pipelines using **Apache Spark** on **AWS EMR**, enabling fast processing of massive datasets and improving performance and scalability.
- Enabled real-time data streaming using **AWS Kinesis** and integrated with **AWS Lambda** for event-driven data ingestion and processing of financial transactions in real-time.
- Integrated **AWS Athena** with **S3,** enabling fast and cost-efficient interactive querying of large datasets, significantly reducing data analysis times.
- Enhanced security by implementing **IAM** policies, data encryption at rest and in transit, and Row-Level Security RLS in **Redshift** to control data access and protect sensitive financial information.
- Deployed machine learning models using **AWS Sage Maker**, applying predictive analytics to identify financial trends and anomalies, supporting business decision-making.
- Automated **ETL** jobs with **AWS Glue ML** Transforms, improving data processing and transforming raw data into business-ready insights.
- Built **CI/CD** pipelines using **AWS Code Pipeline, Jenkins, and Docker**, automating deployment processes for consistent and reliable delivery across environments.

- Monitored **AWS** resources using **AWS CloudWatch**, setting up alarms and dashboards to ensure optimal resource utilization and quick detection of issues across services.
- Managed APIs using **AWS API Gateway**, enabling secure and scalable communication between microservices and external clients.
- Orchestrated and automated complex workflows with AWS Step Functions, integrating multiple AWS services to streamline the end-to-end financial transaction process.
- Integrated **AWS RDS** with data processing workflows for efficient relational database management and optimized performance for large-scale financial data workloads.
- Ensured governance and compliance by utilizing **AWS Service Catalog**, enforcing standardized configurations and access controls across the cloud infrastructure.
- Optimized search and analytics capabilities by integrating Elasticsearch with AWS services for real-time data search and insights.
- Implemented **AWS Lambda** for serverless computing to handle real-time transaction processing, improving scalability while reducing costs.
- Ensured seamless data backup and disaster recovery using **AWS S3** and **Glacier**, implementing data retention and retrieval strategies for compliance and business continuity.
- Google Cloud Dataproc to deploy **Apache Spark** clusters for distributed data processing, improving data processing speeds and reducing processing costs.
- Developed serverless data pipelines with **Google Cloud** Functions, ensuring lightweight, event-driven data processing and cost-effective infrastructure.
- Designed and developed interactive **Tableau dashboards**, providing real-time financial insights by integrating with AWS Redshift and Snowflake, enabling data-driven decision-making for stakeholders.
- Optimized **Tableau** performance by implementing efficient data extracts and aggregation techniques**,** reducing dashboard load times and improving user experience for large-scale financial reporting.
- Implemented **Snowflake** as a **cloud data warehouse**, optimizing storage and query performance for large-scale financial datasets, reducing data retrieval times
- Designed and automated **ELT pipelines** integrating **Snowflake** with **AWS S3 and AWS Glue**, enabling seamless data transformation and ingestion for analytics and reporting.
- Deployed scalable **ETL pipelines** on **AWS Databricks**, leveraging Apache Spark to efficiently process and analyze massive volumes of structured and unstructured financial data.
- Optimized machine learning workflows on **AWS Databricks**, integrating ML models with data lakes and streamlining predictive analytics for fraud detection and financial forecasting.

Environment: AWS S3, EC2, Redshift, Redis, EMR, Kinesis, Lambda, Step Functions, Glue, Athena, DynamoDB, Elasticsearch, Service Catalog, CloudWatch, IAM, Sage Maker, RDS, API Gateway, AWS CloudWatch, AWS Glue ML Transforms, Amazon Athena. Big Query, Dataproc, Pub/Sub, Cloud Storage, AI Platform, Data Studio, Cloud Functions, Deployment Manager, Cloud Endpoints, Cloud IAM, Bigtable, Kubernetes Engine, Cloud Composer, Tableau, AWS Databricks, Snowflake.

Vitech Asia Pvt Ltd **April 2020 – July 2022**
Azure Data Engineer
Responsibilities:

- Designed and implemented robust data pipelines using **Azure Data Factory ADF** to orchestrate the seamless movement of data from on-premises systems to **Azure SQL Database** and **Azure Synapse Analytics**, enabling real-time analytics and decision-making.
- **Developed** real-time data streaming solutions Azure Stream Analytics **and** Apache Kafka**, processing high-volume telemetry data from IoT devices and ingesting it into** Azure Data Lake Storage **for scalable storage and analysis.**
- Optimized **ETL workflows** in **Azure Data Factory**, automating data movement and transformation processes to improve data flow efficiency across various cloud services and systems.
- **Developed data models and views** in **Azure SQL Database** and **Azure Synapse Analytics**, supporting complex analytical workloads and telecom operations for performance and usage tracking.

- Created **interactive dashboards and reports** using **Power BI**, providing telecom operators with insights into usage trends, customer behavior, service performance, and network health.
- **Azure Databricks** to build scalable data processing pipelines and machine learning models for predictive analytics, enabling telecom operators to forecast customer needs and optimize network resources.
- Implemented data security measures using **Azure Key Vault** to securely store sensitive information, and Role-Based Access Control RBAC to manage access permissions, ensuring data protection and compliance with regulatory standards.
- Integrated **Azure Data** Explorer to perform high-performance querying and analytics on large-scale structured and semi-structured data, enabling rapid insights into network logs and performance metrics.
- Migrated legacy on-premises data systems to **Azure Data Lake Storage** and **Azure Synapse**, modernizing infrastructure and improving data accessibility, performance, and scalability while reducing operational costs.
- Developed and monitored cloud applications with **Azure Monitor**, tracking performance metrics and operational health, automating alerts to proactively address system performance issues.
- Orchestrated big data workflows using **Azure HDInsight**, enabling the deployment of Hadoop and Spark-based clusters for high-volume data processing and analysis, improving scalability and performance for telecom operations.
- Automated **CI/CD workflows** using **Azure DevOps**, streamlining development pipelines and ensuring consistent, efficient deployment of data pipelines and analytical models.
- Integrated **Azure Data Explorer** with real-time data streams from **Azure IoT Hub**, enabling operators to monitor IoT device data and network performance in real-time for better decision-making.
- **Optimized cloud storage solutions** using **Azure Blob Storage** for structured and unstructured data, ensuring cost-effective data management and retrieval across the telecom ecosystem.
- **Azure Data Lake** to store large volumes of unstructured data, supporting advanced analytics and machine learning workflows for telecom services like predictive maintenance and usage forecasting.
- Supported the migration of business-critical data from on-premises systems to cloud-based **Azure Synapse** and **Azure Data Lake Storage**, improving analytics capabilities while optimizing operational costs.
- Collaborated with data scientists and analysts to prepare and clean large datasets using **Azure Databricks**, ensuring high-quality inputs for predictive models and advanced analytics.
- Orchestrated big data workflows using **Azure HDInsight** and **Apache Airflow**, enabling the deployment of **Hadoop** and **Spark-based clusters** for high-volume data processing and analysis, improving scalability and performance for telecom operations.
- Enabled cross-platform data integration by using **Azure Data Factory**, **Apache Airflow**, and **Snowflake** to connect on-premises systems with cloud-based applications, automating data movement and transformation.
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Environment: **Azure Data Lake**, **Azure Data Factory**, **Azure Databricks**, **Azure SQL Database**, **Azure Synapse**, **Azure Analysis Services**, **Azure Blob Storage**, **Azure Monitor**, **Azure Key Vault**, **Azure HDInsight**, **Azure Data Explorer**, **Power BI**, **Kafka**, **Azure DevOps**, **Airflow, Snowflake.**

Tata Consultancy Services **Feb 2019 – April 2020**
Data Engineer
Responsibilities:

- Designed and developed robust **ETL pipelines** using **Apache Hive**, **Hadoop**, and **Python** to efficiently process large volumes of structured and unstructured data, ensuring seamless integration into centralized data warehouses for downstream analytics.
- Implemented batch and real-time data processing solutions by utilizing **Apache Kafka** for real-time ingestion, enabling continuous updates to be reporting systems and providing real-time insights into sales, inventory, and customer behaviors.
- Managed **data lake** using **Hadoop**, ensuring scalable and cost-effective storage of structured and unstructured retail data, which allowed analysts and data scientists to perform advanced analytics and reporting.
- Optimized **SQL queries** in **Hive** and **Postgres**, reducing query execution times significantly, improving the performance of reporting systems, and providing near-instantaneous access to large datasets.

- Streamlined **ETL workflows** using **Apache Spark** and **Hive**, transforming vast amounts of retail data efficiently while maintaining high performance and scalability.
- Developed interactive dashboards using **Power BI** empowering business users with real-time key performance indicators KPIs and actionable insights into sales trends, customer behaviors, and inventory health.
- Integrated third-party data's sources via **APIs** into the data lake, enriching the retail dataset and improving customer insights, enabling more accurate demand forecasting and targeted marketing.
- Implemented error handling mechanisms and data validation processes in **ETL pipelines** to ensure data integrity, reduce failures, and improve the overall quality of data.
- Automated **ETL pipeline** orchestration with **Apache Airflow**, handling scheduling, retries, and notifications to ensure smooth, error-free data processing and operational efficiency.
- Managed large-scale data transformations by leveraging **Hadoop** and **Apache Spark**, processing petabytes of data and providing transformed datasets to support complex analytics.
- Ensured data security and compliance by implementing access controls and data encryption in **ETL** processes, guaranteeing that sensitive information was properly protected throughout the data pipeline.
- Worked closely with data scientists and analysts to prepare and clean data for **machine learning** models, ensuring high-quality inputs and enhancing the predictive capabilities of the organization.
- Developed custom **ETL solutions** to handle specific business requirements, such as integrating data from IoT devices, social media platforms, and other third-party sources for more comprehensive reporting and analysis.
- Improved data accessibility by implementing cataloging solutions and data warehouses, allowing users to quickly access and query both raw and transformed datasets using tools like **Hive** and **Postgres**.
- Ensured efficient data transformation by applying partitioning, bucketing, and indexing techniques in **Hive** and **Spark** to optimize read and write operations across large datasets.
- Designed and implemented **ETL** workflows using **Informatica** to efficiently extract, transform, and load data from multiple sources into data warehouses, improving data processing efficiency and ensuring high data quality.
- Optimized and automated **ETL** processes using **Informatica PowerCenter**, reducing processing time by 40% and enhancing scalability for handling large datasets across retail operations.

Environment: **Apache Hive, Apache Hadoop, Apache Kafka, Apache Spark**, **Python, PostgreSQL, Power BI, AWS S3, Apache Airflow, ETL Pipelines, Data Lakes, Data Security and Compliance, Data Archiving, Distributed Computing, Informatica ETL.**

Bhavishyath Digital Technologies Pvt. Ltd **Jan 2017 – Feb 2019**
Python Developer
Responsibilities:

- Developed simple **ETL workflows** using **Python** to process and transform healthcare data, ensuring clean and structured datasets for reporting.
- Created small **Python scripts** to automate data cleaning and transformation tasks, improving data quality and consistency for reporting purposes.
- Contributed to the development of real-time data ingestion processes, using **Python** and **Apache Kafka** to streamline the flow of data from medical devices.
- Used **Python libraries** like **Pandas** to clean and manipulate small to medium-sized healthcare datasets, making them ready for further analysis.
- Assisted with **API** integration by writing simple **Python** scripts to connect with healthcare systems and pull relevant data for processing and reporting.
- Wrote **Python code** to help automate basic ETL tasks, such as data extraction from flat files and transformation into structured formats for storage.
- Collaborated with data scientists to prepare data for machine learning models by cleaning and structuring datasets using **Python**.
- Developed **Python scripts** for generating simple reports and dashboards, making it easier for healthcare professionals to track patient trends.
- Helped maintain **data pipelines** by writing and executing **Python scripts** for scheduled ETL tasks, ensuring smooth data flow.

- Assisted in the development of error-handling mechanisms within **Python ETL** workflows, ensuring data integrity and reducing failures.
- Learned and followed best practices for HIPAA compliance in data handling, using **Python** to implement basic encryption methods for sensitive healthcare data.
- Collaborated with team members to implement monitoring scripts for ETL processes, using Python to ensure that data flows smoothly through the pipeline.
- Assisted with the development of data validation processes using **Python** to ensure the accuracy and consistency of healthcare data.
- Developed small-scale **Python** tools to automate repetitive tasks, improving operational efficiency and reducing manual efforts.
- Worked on improving code quality by refactoring **Python scripts** for better performance and readability, under the guidance of senior developers.
- Learned to use **version control tools** like **Git** to collaborate on Python code with team members, ensuring proper code management and tracking changes.

Environment: Python, Apache Kafka, Pandas, NumPy, EMR, API.

Education:

- Masters in the Information Technology from Arkansas Tech University, Russellville, AR – 2022
  GPA:4.0